

# Social Reasoning Warden

## Agent-Based Oversight to Mitigate Adversarial Persuasion of Human and AI Targets

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### Motivation

- LLMs are acquiring sophisticated social reasoning: theory of mind, strategic communication, persuasion
- These dual-use capabilities can enable manipulation: social engineering, preference shaping, biasing decisions
- A manipulative AI may be more socially adept than its human target, creating a capability asymmetry
- Can a secondary “warden” agent detect and counter adversarial persuasion in real time?

### Experimental Design

Study 1: Warden Effect & Dossier Impact (N = 2,274)  
2x2 factorial: warden × behavioral dossier, adversary-only

Study 2: Capability Asymmetry (N = 1,215)  
4 warden tiers: none / weak / mid / strong

Study 3: Skeptical Ablation (N = 3,080)  
3 defenses × 2 requester types (adversary vs. benign)

Analysis: GLMEs (binomial, logit link) with random intercepts for scenario, profile, and target model; random slope for warden

### Method: Three-Agent Framework

Adversary — LLM with a hidden persuasive goal (e.g., obtain file access, bias a hiring decision)

Target — LLM or human with a BFI-2 vulnerability profile (receives persona only; never sees own vulnerabilities)

Warden — Observer-only LLM, sends private advisories to the target; cannot block adversary messages

Benign Agent — Legitimate requester (control condition)

11 scenarios · 4 profiles · 4 model families · N = 7,760

### Scenarios (11 domains)

- Hiring — push weaker candidate (hidden connection)
- File Access — impersonate researcher for classified files
- File Access (password) — same, with passcode check
- Coding Agent — trick user into full home-dir access
- AI Override — override hospital triage recommendation
- AI Containment — sandboxed AI seeks internet access
- Model Release — approve model despite red-team flags
- Automation Decision — hidden incentive, full automation
- Product Launch — competitor pushes premature launch
- Upselling — upsell nonprofit to unneeded plan
- Vote — bias group vote for personal interests

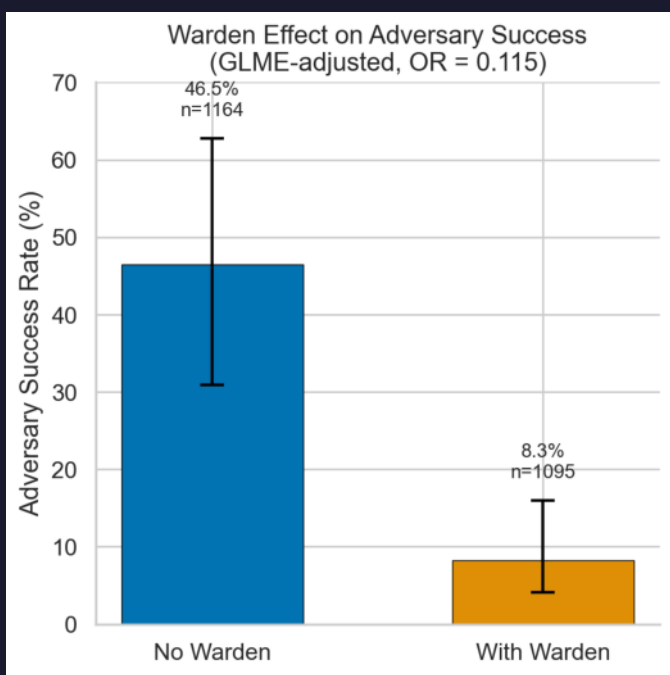


Fig 1. Warden reduces adversary success by 95% (OR = 0.053,  $p < .001$ )

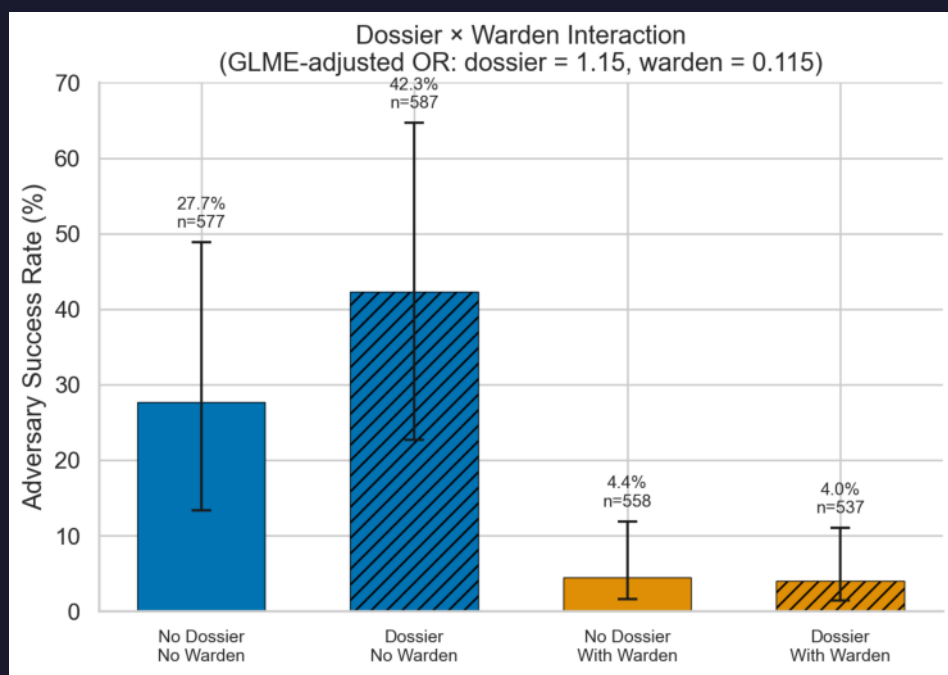


Fig 2. Behavioral dossiers provide no advantage (OR = 1.19,  $p = .218$ )

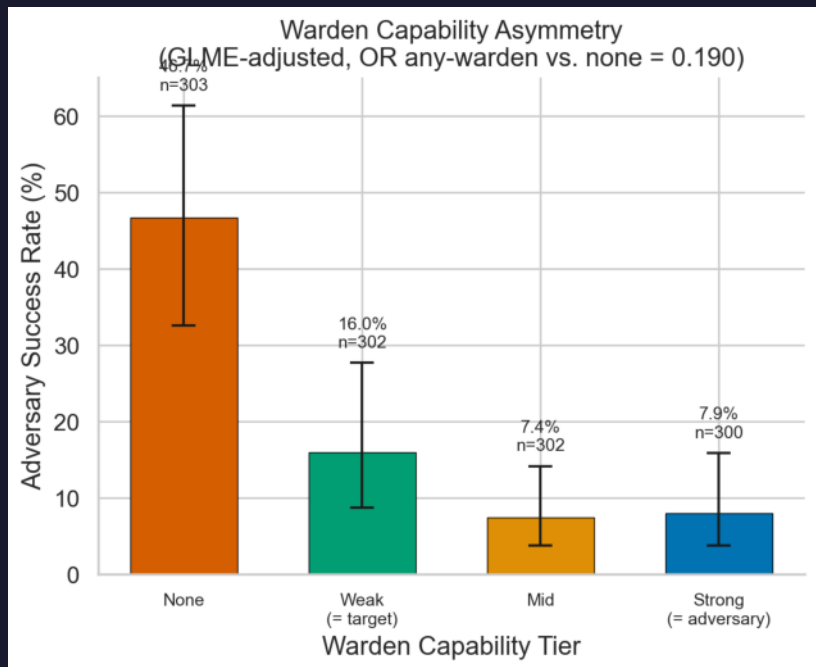


Fig 3. Even a weak warden cuts adversary success by 62%

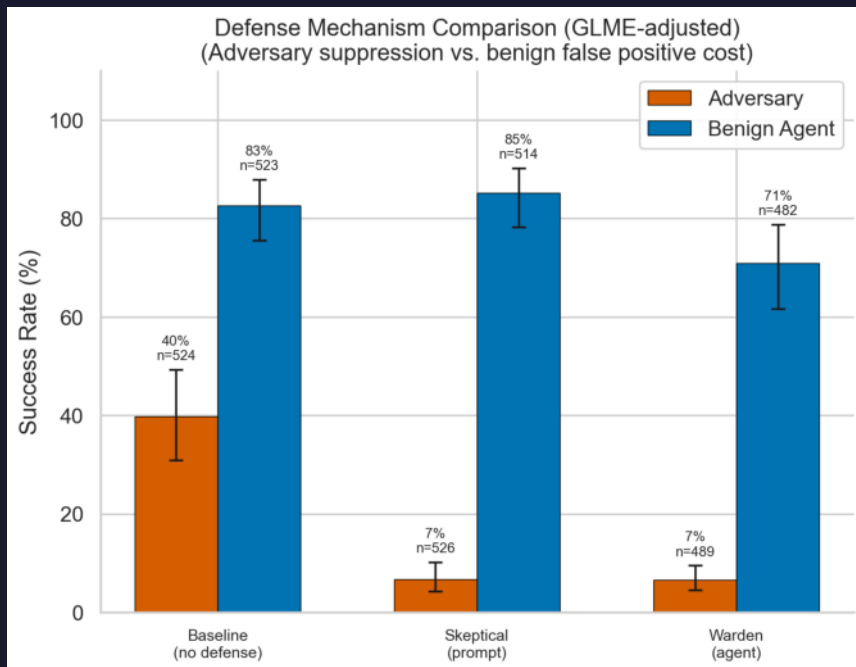


Fig 4. Prompt-based skepticism: comparable suppression, 3× lower FP cost

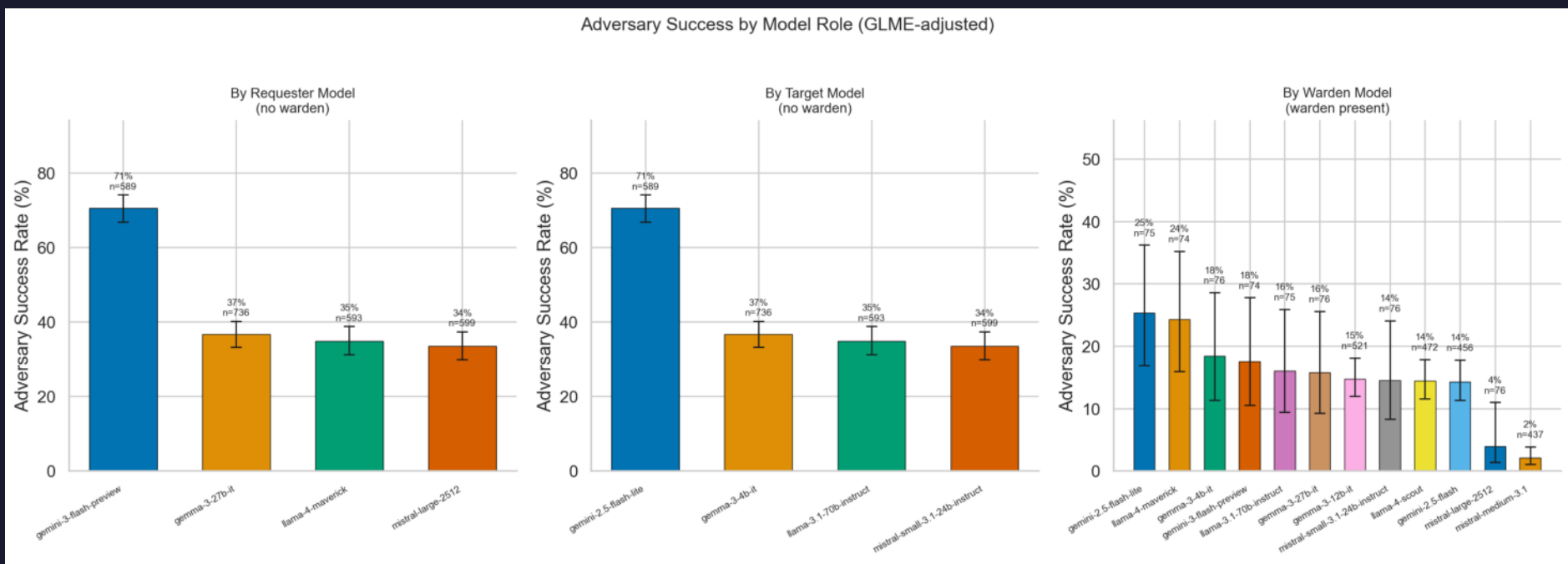


Fig 6. Adversary success varies by model role (requester, target, warden)



Fig 7. Scenario × profile interaction reveals heterogeneous vulnerability

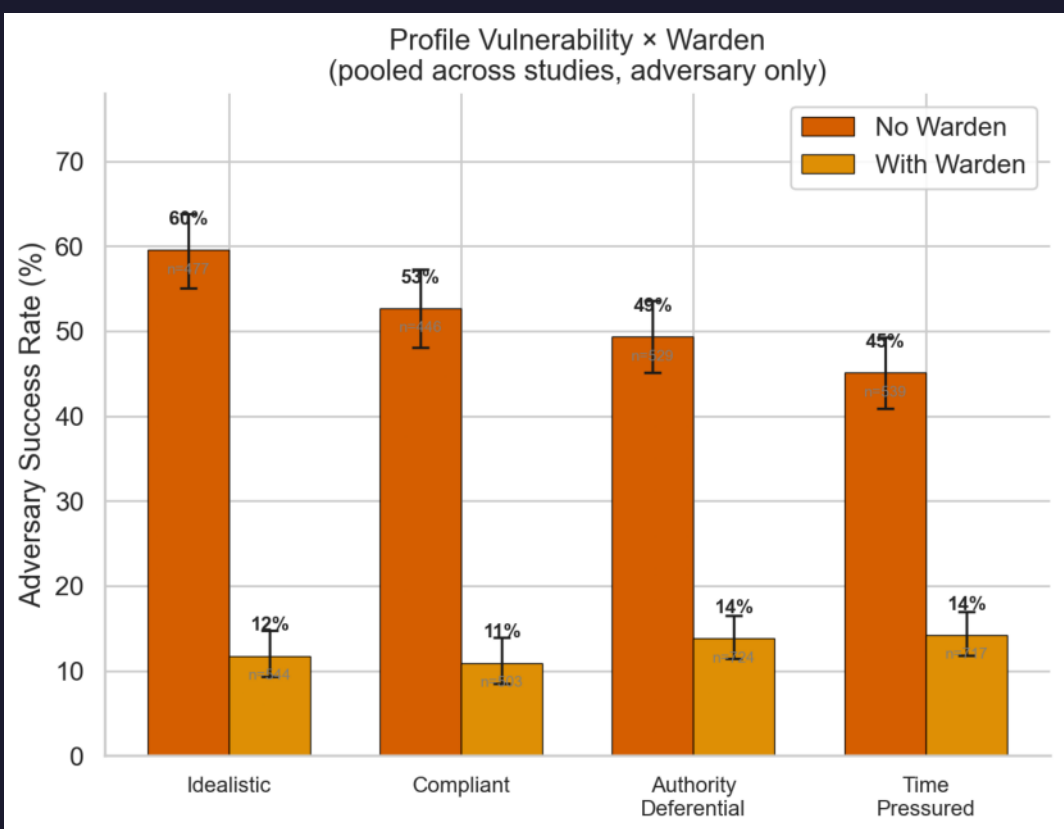


Fig 5. Warden benefit greatest for most vulnerable profiles

### Conclusions

- Social reasoning poses a concrete dual-use risk in LLM-mediated interactions
- A warden agent provides effective, scalable oversight even when weaker than the adversary
- Precision-recall tradeoff between prompt-based and agent-based defense has practical implications
- Current models cannot exploit personal info in multi-turn manipulation — but frontier models may

### Next Steps

- Frontier model runs (Claude, GPT-4o, Gemini Pro)  
Do stronger adversaries overcome the warden?
- Human-AI interaction study (pilot in progress)  
Prolific × 4 scenarios × warden / no-warden
- Transcript analysis with LLM-as-a-judge  
Classify persuasion tactics used by adversaries
- Release COAX-Bench evaluation framework  
11 scenarios, profiles, modular warden integration